

Week 1, Lecture 1 Handout: Web Foundations

Web Basics: How the Internet Works

The internet is a global network of connected computers that communicate with each other using a specific set of rules. When you visit a website, a structured conversation happens between two entities:

- **The Client:** The device and software (like Google Chrome, Microsoft Edge, or Safari) you use to browse the internet. Its job is to request files and turn code into a visual interface.
- **The Server:** A high-powered, remote computer that stores all the website's files (HTML, CSS, JavaScript, and images) and waits for clients to ask for them.
- **The Domain Name System (DNS):** The phonebook of the internet. Computers use numerical IDs called **IP Addresses** (e.g., `192.0.2.1`) to find each other. Because humans cannot easily remember long strings of numbers, the DNS translates domain names (like `google.com`) into the exact IP address of the server hosting that website.
- **The Request-Response Cycle:**
 1. You type a URL into your browser.
 2. The browser asks the DNS for the server's IP address.
 3. The browser sends an **HTTP Request** across the internet to that server.
 4. The server processes the request and sends back an **HTTP Response** containing the website's code files.
 5. The browser processes this code line-by-line, creating an interactive blueprint called the **DOM (Document Object Model)**, and paints the pixels on your screen.

What is Web Development?

Web Development is the process of building, creating, and maintaining websites and web applications that run over the internet. It ranges from writing simple, static text pages using plain markup to constructing highly secure, data-driven platforms and social networks.

It is split into two primary areas:

- **Frontend Development:** Building the visual layer that users see, click, and interact with using HTML, CSS, and JavaScript.
- **Backend Development:** Managing the invisible infrastructure, servers, and databases that process and store information behind the scenes.

The Professional Developer Environment

To write clean, industry-standard code, software developers use a professional local toolkit rather than basic text editors:

- **Integrated Development Environment (IDE) - VS Code:** The global standard text editor optimized for writing code. It features color-coded text, syntax highlighting, and predictive typing.
 - **Prettier:** A built-in code formatting extension that automatically keeps your code clean, nested, and legible the moment you save a file.
 - **Live Server:** A local development extension that launches a private server on your machine, instantly refreshing your browser window whenever you hit save.
 - **Browser DevTools:** A built-in suite of debugging tools accessible by right-clicking any webpage and choosing **Inspect**. It allows you to peer inside the live DOM tree via the **Elements** tab and view system logs or errors via the **Console** tab.
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Document Initialization & Core Text Tags

Every professional HTML document requires a specific foundational skeleton to communicate with the browser correctly. Below are the essential starting components and basic text elements:

1. The Core Skeletons & Metadata Tags

- **<!DOCTYPE html>**
The document type declaration. It must be the very first line of code at the top of your file to inform the browser that the page is written in modern HTML5.
- **<html>**
The root element that wraps all the content on your entire page. Every other tag must be nested inside this element.
- **<head>**
A container for machine-readable information (metadata) about the page that does not display to the user directly, such as character sets, configuration settings, and external stylesheets.
- **<title>**
Nested inside the **<head>**, this tag sets the name of the webpage as it appears on the browser tab and in search engine results.
- **<body>**
The main container for all the visible content on your webpage. Everything written inside this tag—text, images, links, and forms—is what the user actually sees on screen.

2. The Text Elements

- **<h1> to <h6> (Heading Tags)**
Structural section headings used to establish visual and logical hierarchy. **<h1>** represents the single most important heading on the page, down to **<h6>** which represents the lowest level sub-heading.
 - **<p> (Paragraph Tag)**
The foundational block for body text, used to organize generic blocks of sentences and plain copy with default margin spacing.
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Freelancing Tip of the Day: The Power of Communication

Your technical skills get you noticed, but your communication skills get you hired. In the freelancing landscape, clients are often thousands of miles away across different time zones. Clear, prompt, and proactive communication builds instant trust.

Always explain what you are building using plain, client-friendly terms instead of dense developer jargon. Replying quickly to messages, managing delivery expectations honestly, and clearly defining the project scope before writing code prevents misunderstandings and turns single-order clients into high-paying, long-term partners.

To fast-track your client management skills, review these comprehensive professional strategies:

 [15 Communication Tips That Can Make You a Successful Freelancer](#)

Homework Assignment: Your First Web Page

Objective: Create your very first webpage using the exact tags learned in class today.

Task Instructions

1. Open **VS Code** and create a new file named `index.html`.
 2. Set up your basic skeleton using `<!DOCTYPE html>`, `<html>`, `<head>`, `<title>`, and `<body>`.
 3. Set the browser tab text (`<title>`) to your name.
 4. Inside the `<body>`, write:
 1. Your name using an `<h1>` tag.
 2. Your current or target job title using an `<h2>` tag.
 3. A short, one-sentence bio about yourself using a `<p>` tag.
 5. Save your file and open it using **Live Server** to see your work live in the browser!
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